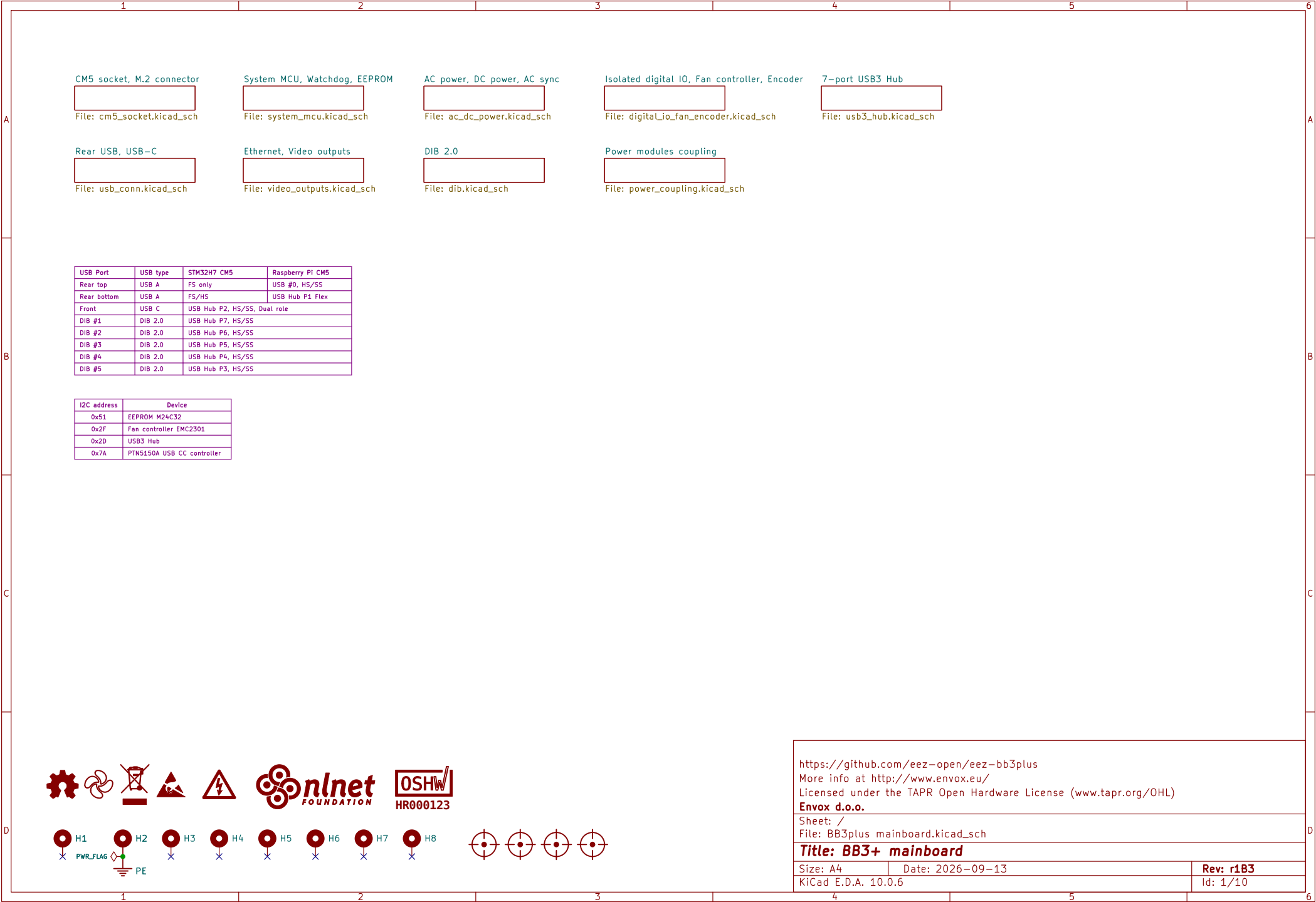
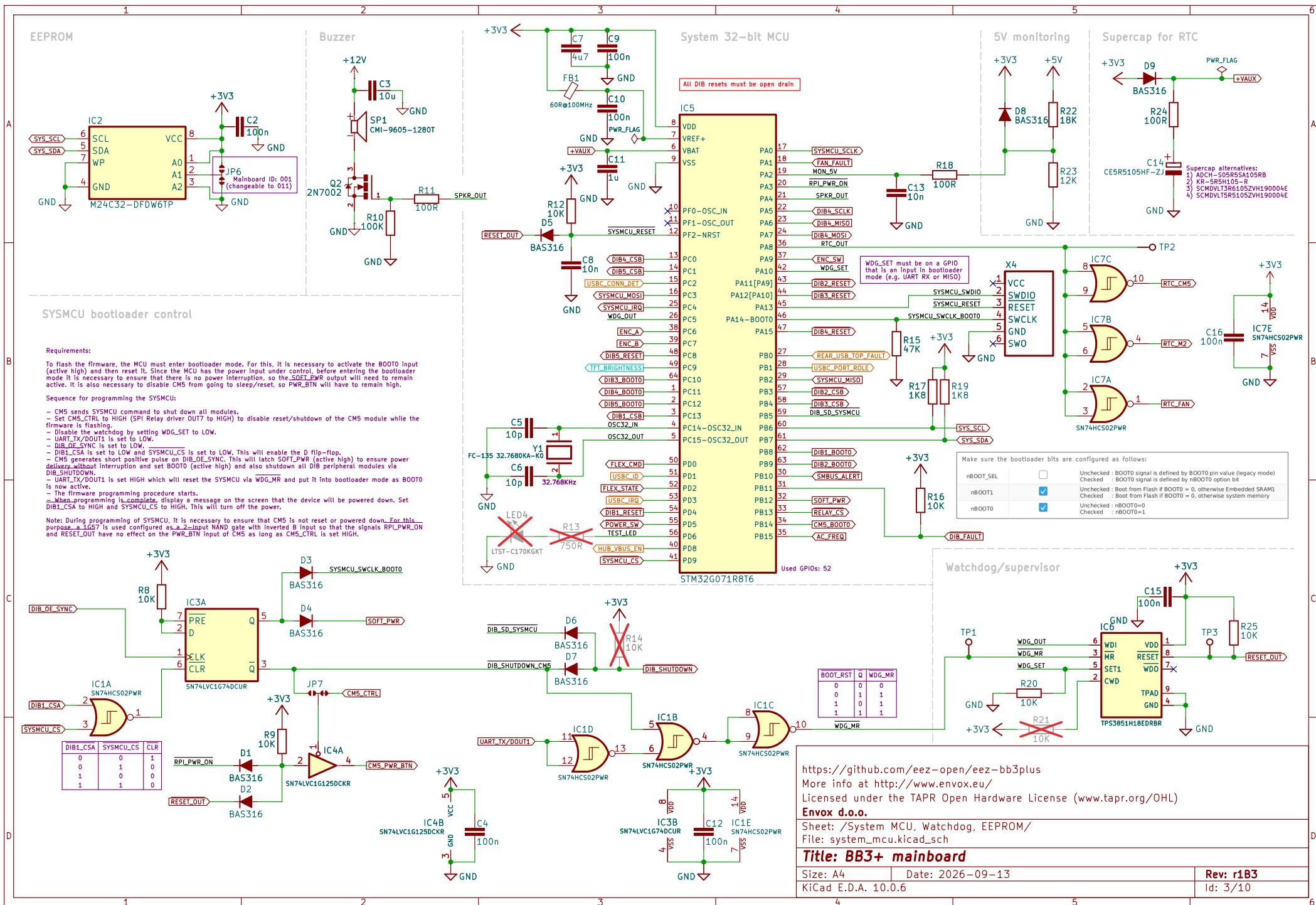








<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /System MCU, Watchdog, EEPROM/

File: system_mcu.kicad_sch

Title: BB3+ mainboard

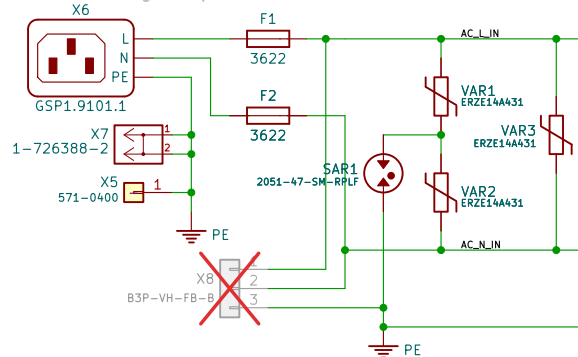
Size: A4 Date: 2026-09-13

KiCad E.D.A. 10.0.6

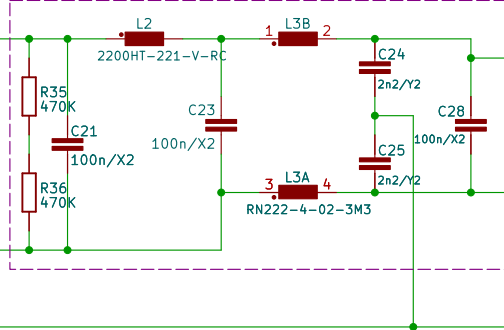
Rev: r1B3

Id: 3/10

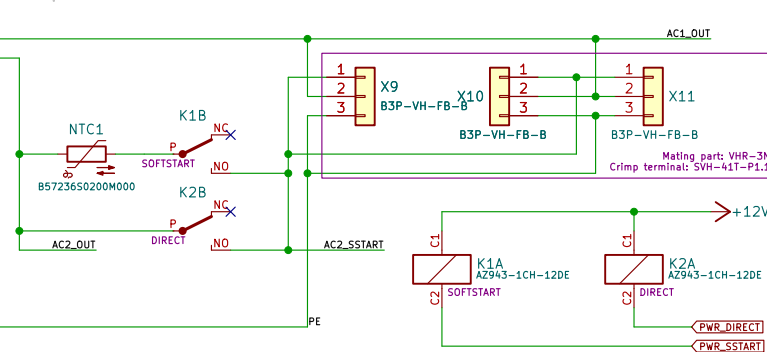
AC inlet, filtering and protection



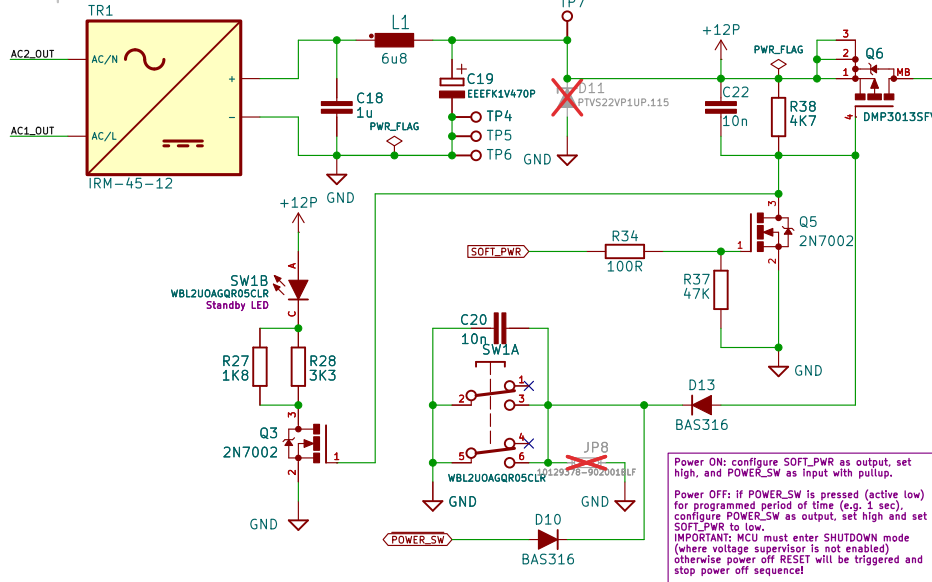
EMI compliance filter



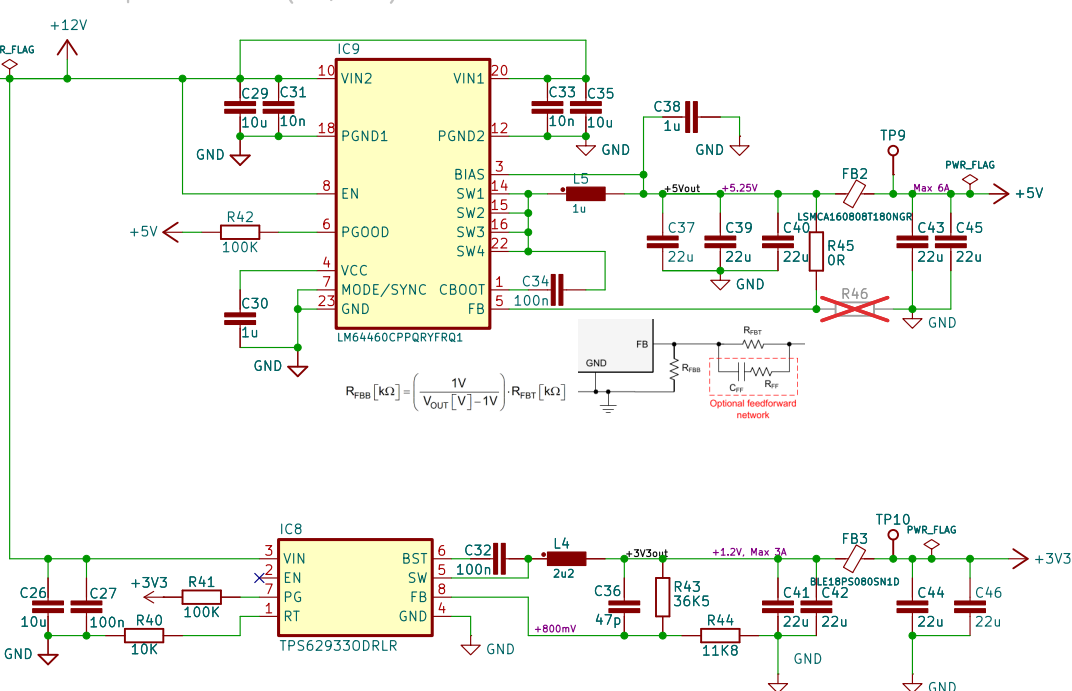
AC power control, Inrush current limiter



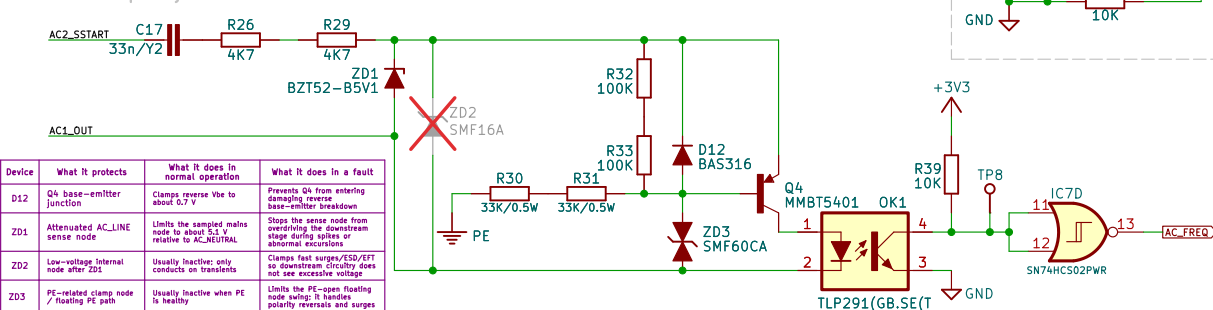
Soft power switch



DC-DC stepdown converters (+5V, +3V3)



AC Mains frequency and PE fault detection



Device	What it protects	What it does in normal operation	What it does in a fault
D12	Q4 base-emitter junction		Prevents Q4 from entering damaging reverse base-emitter breakdown
ZD1	Attenuated AC_LINE sense node	Limits the sampled mains node to about 5.1 V relative to AC_NEUTRAL	Stops the sense node from overdriving the downstream stage during spikes or abnormal excursions
ZD2	Low-voltage internal node after ZD1	Usually inactive; only conducts on transients	Clamps fast surges/ESD/EFT so downstream circuitry does not see excessive voltage
ZD3	PE-related clamp node / floating PE path	Usually inactive when PE is healthy	Limits the PE-open floating node swing; it handles polarity reversals and surges

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /AC power, DC power, AC sync/

File: ac_dc_power.kicad_sch

Title: BB3+ mainboard

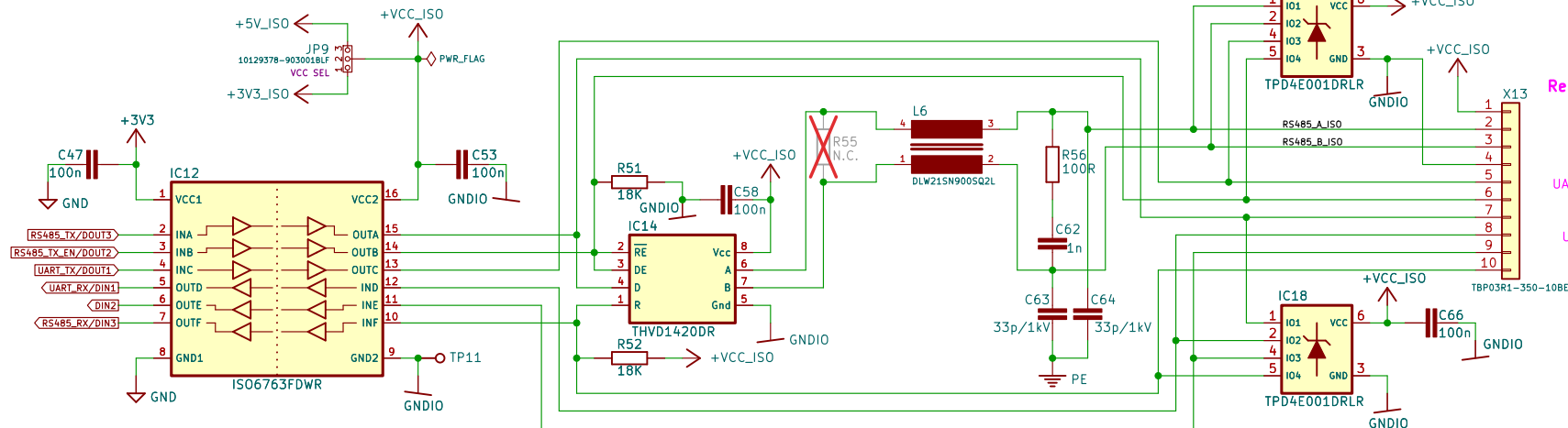
Size: A4 Date: 2026-09-13

KiCad E.D.A. 10.0.6

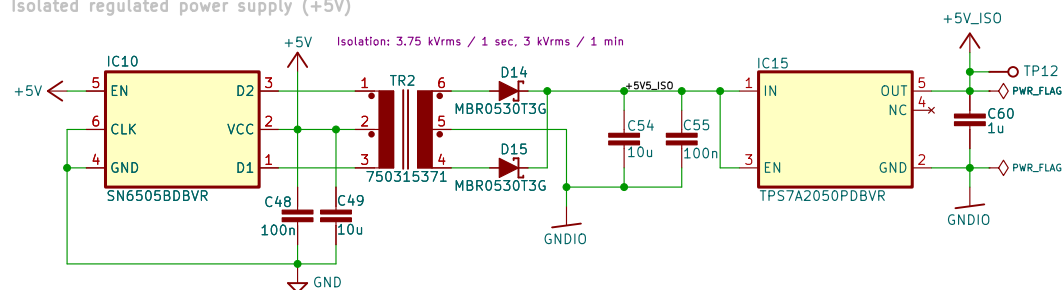
Rev: r1B3

Id: 4/10

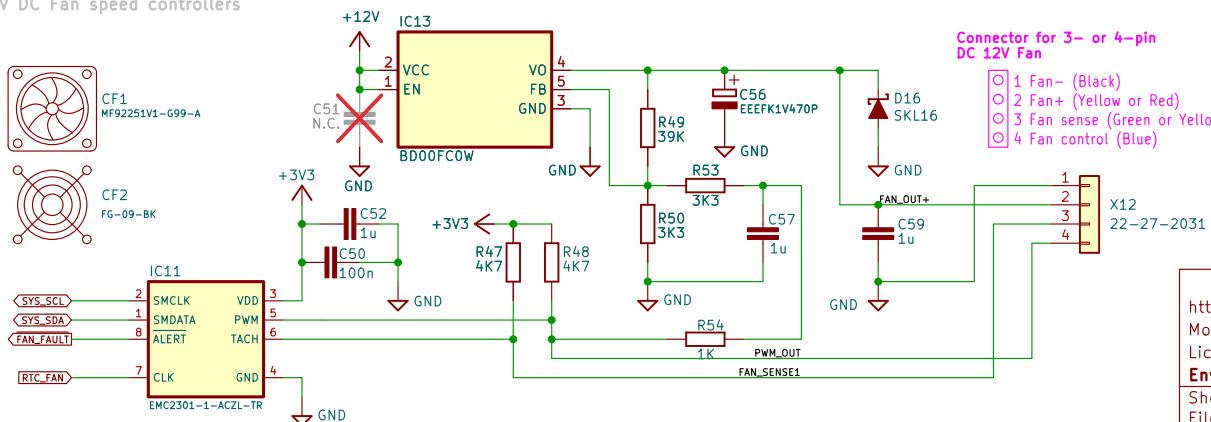
Isolated digital I/O & RS485 transceiver (half-duplex)



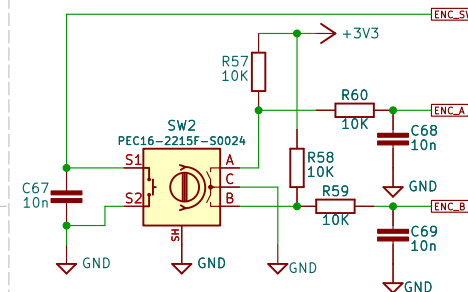
Isolated regulated power supply (+5V)



12V DC Fan speed controllers



Encoder with switch



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Envox d.o.o.

Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

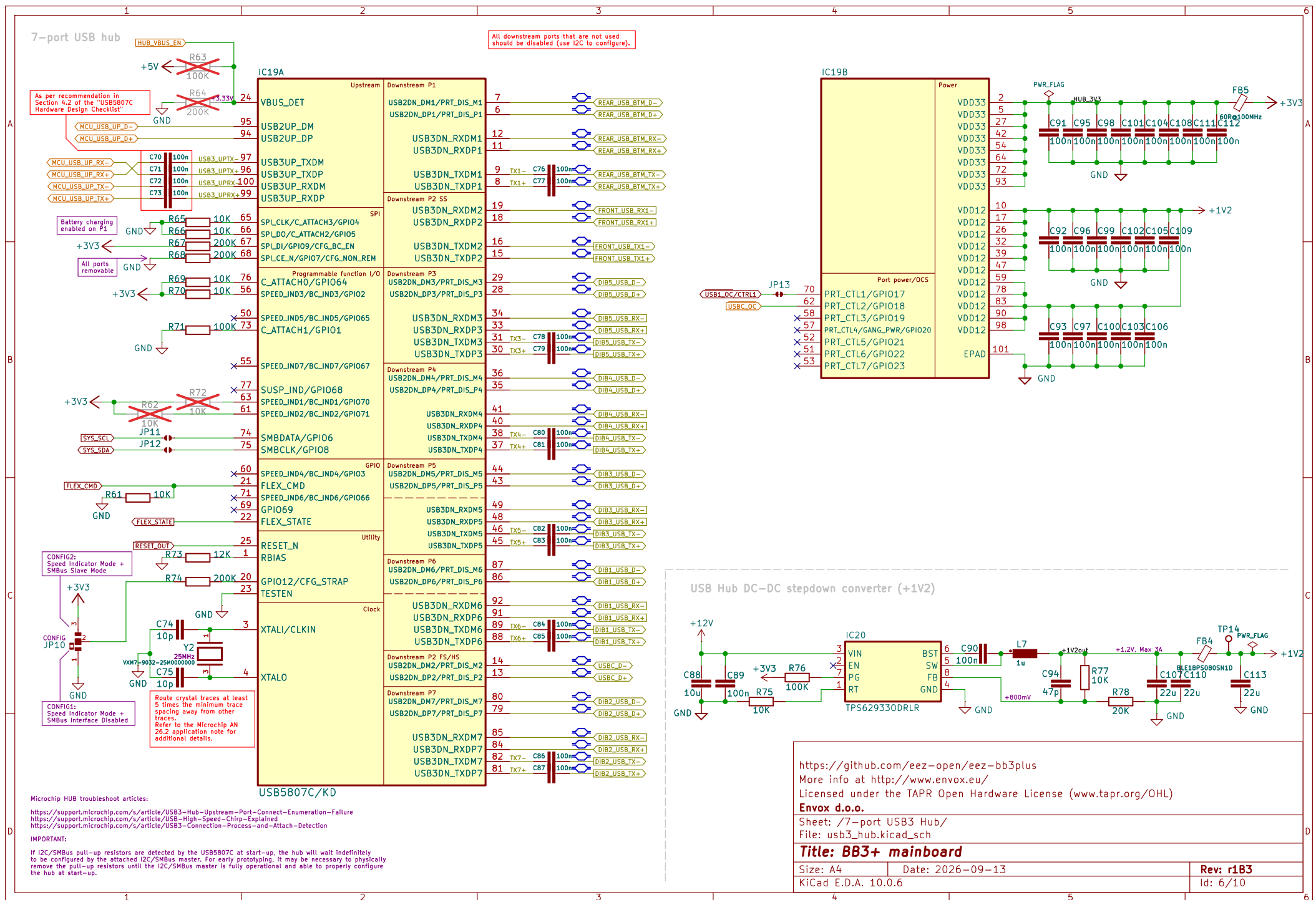
Title: BB3+ mainboard

Size: A4 Date: 2026-09-13

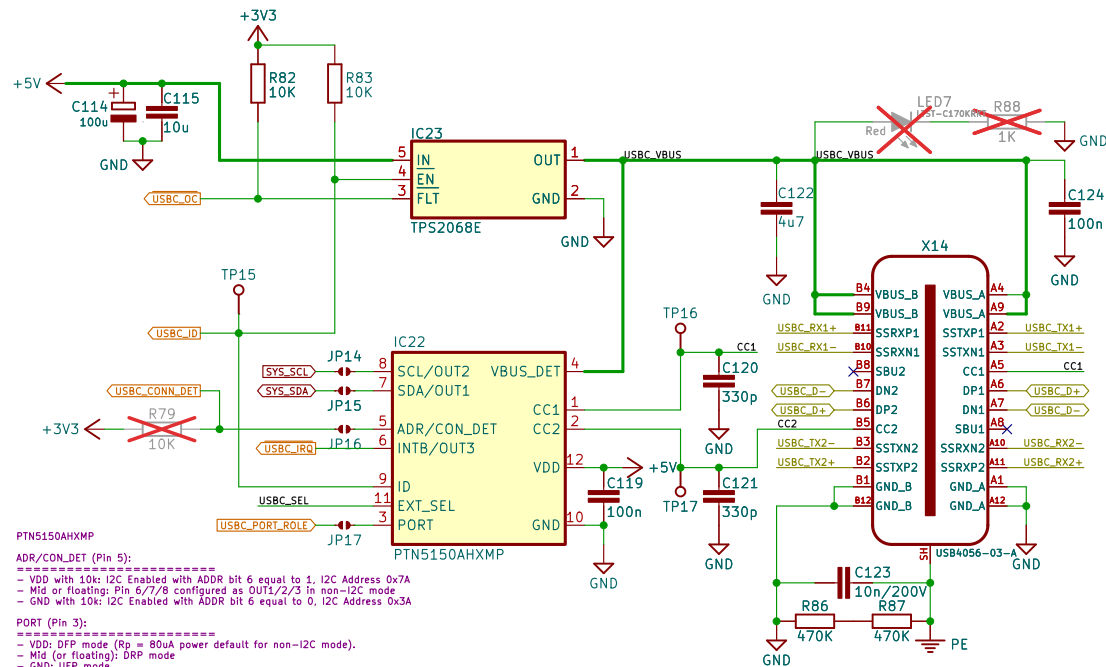
KiCad E.D.A. 10.0.6

Rev: r1B3

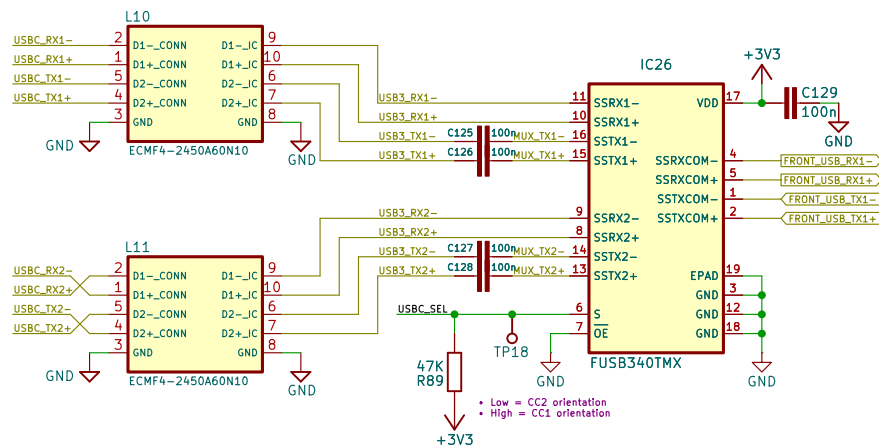
Id: 5/10



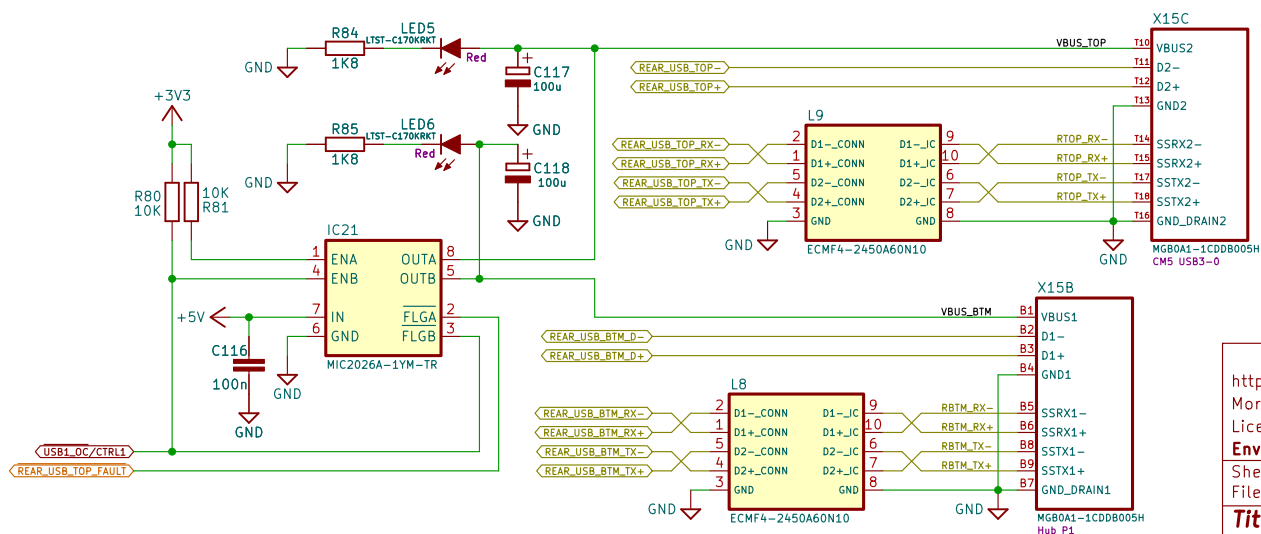
USB-C 3.0 front port (source max 1.5A)



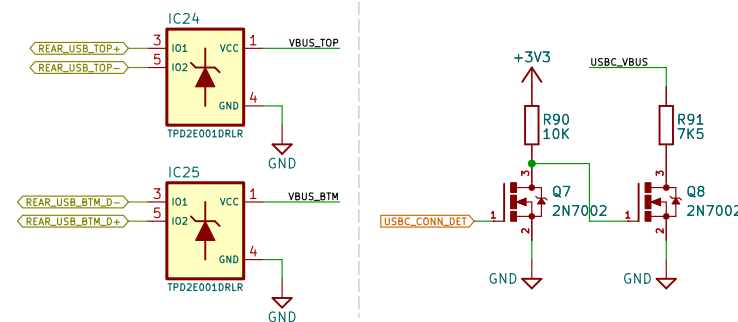
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



VBus Discharge Circuit



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Sheet: /Rear USB, USB-C/

File: usb_conn.kicad_sch

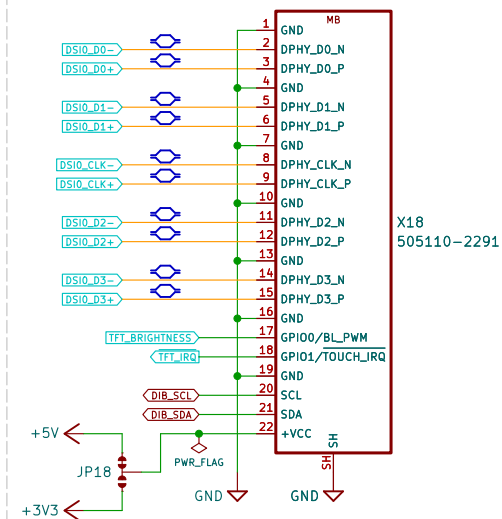
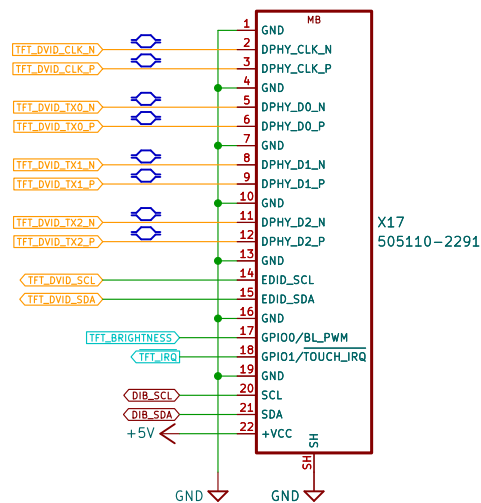
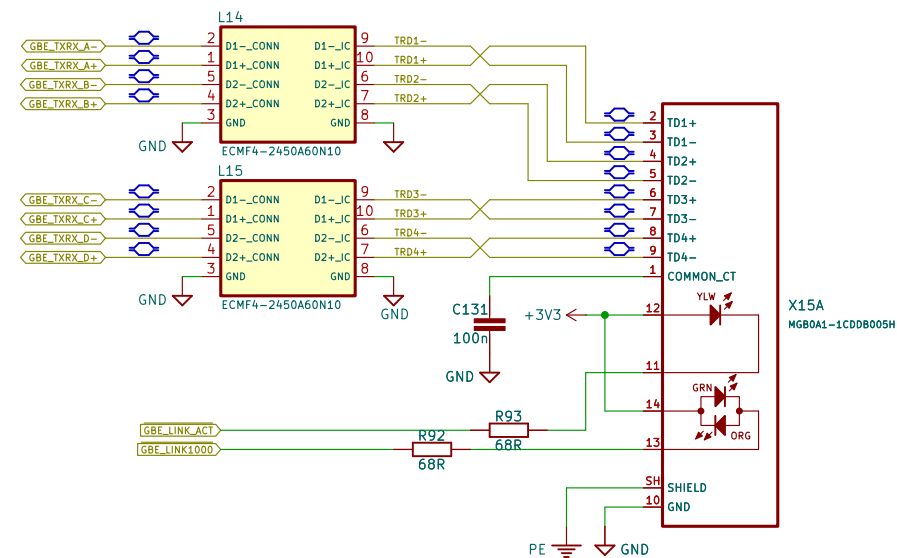
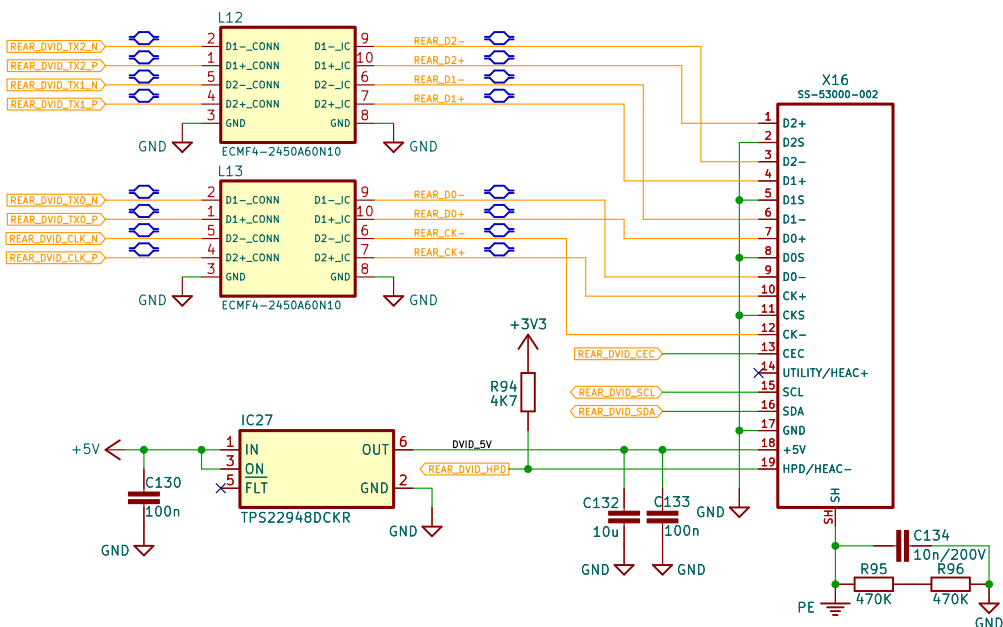
Title: BB3+ mainboard

Size: A4 Date: 2026-09-13

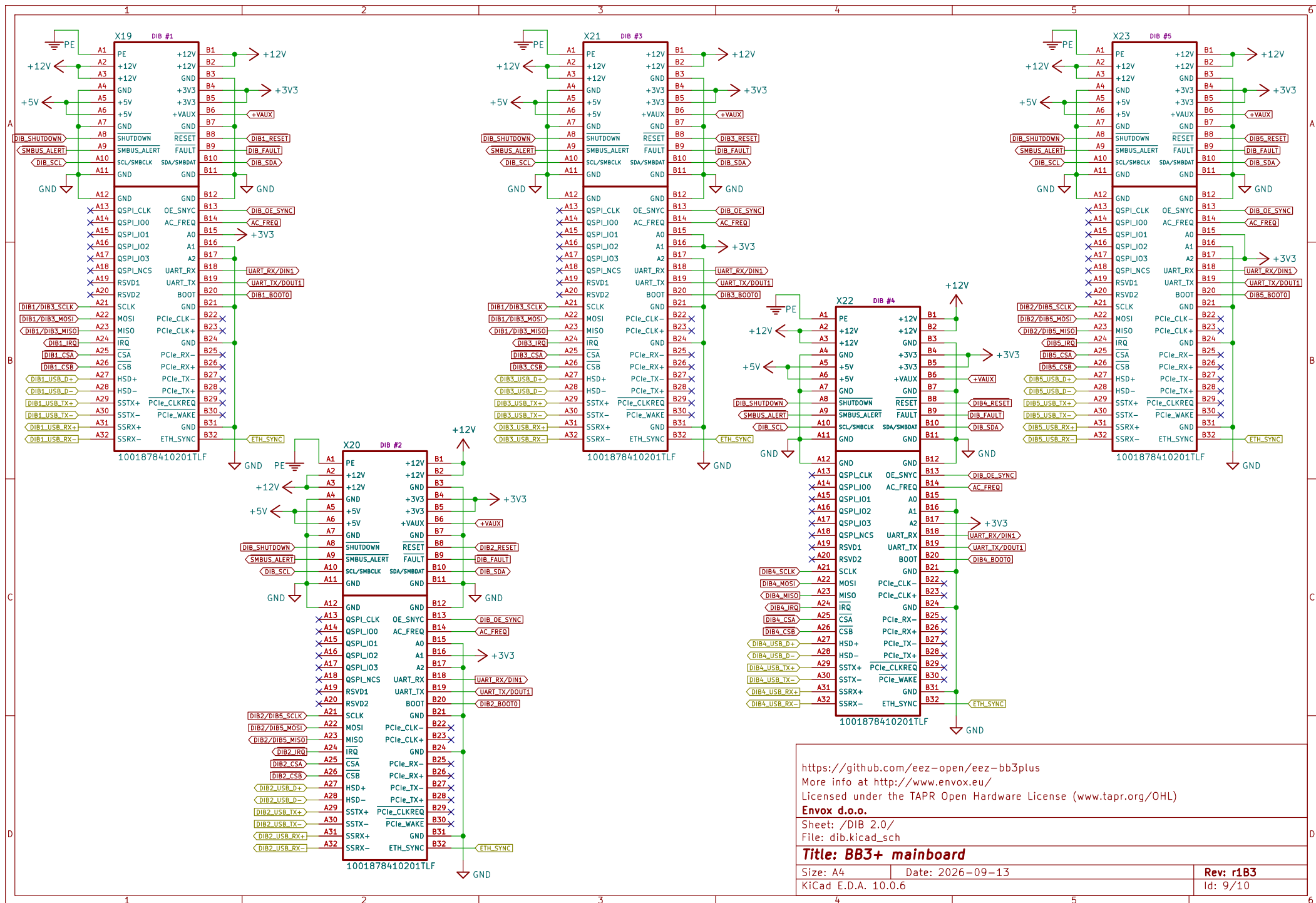
KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 7/10



Id: 8/10



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Envox d.o.o.

Sheet: /DIB 2.0/

File: dib.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-09-13

KiCad E.D.A. 10.0.6

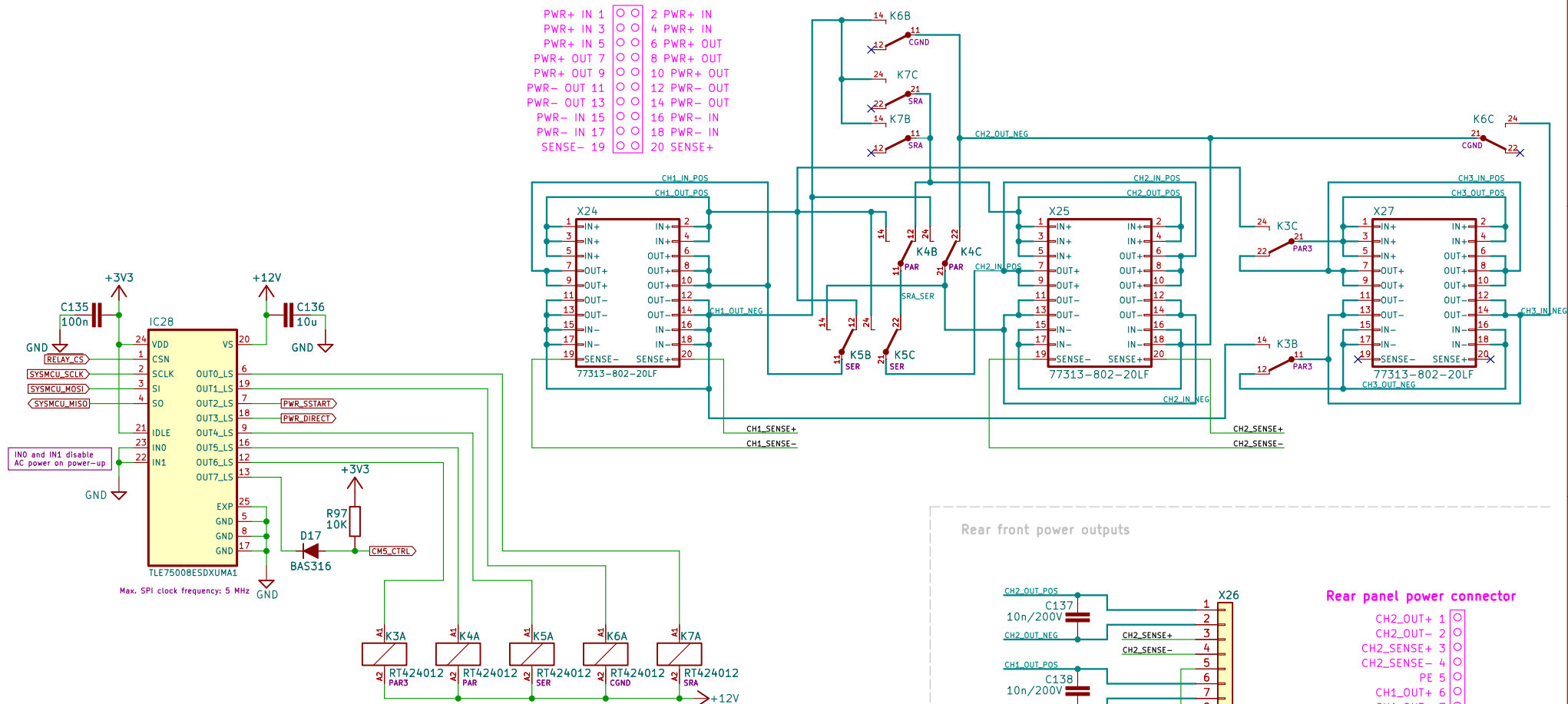
Rev: r1B3

Id: 9/10

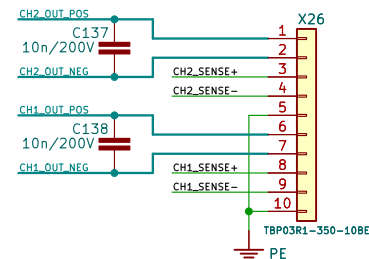
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	☐	☐	2 PWR+ IN
PWR+ IN 3	☐	☐	4 PWR+ IN
PWR+ IN 5	☐	☐	6 PWR+ OUT
PWR+ OUT 7	☐	☐	8 PWR+ OUT
PWR+ OUT 9	☐	☐	10 PWR+ OUT
PWR- OUT 11	☐	☐	12 PWR- OUT
PWR- OUT 13	☐	☐	14 PWR- OUT
PWR- IN 15	☐	☐	16 PWR- IN
PWR- IN 17	☐	☐	18 PWR- IN
SENSE- 19	☐	☐	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	○
CH2_OUT- 2	○
CH2_SENSE+ 3	○
CH2_SENSE- 4	○
PE 5	○
CH1_OUT+ 6	○
CH1_OUT- 7	○
CH1_SENSE+ 8	○
CH1_SENSE- 9	○
PE 10	○

<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-09-13
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KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 10/10